

FLIPKART PRINTER MARKET ANALYSIS

Web Scraping & E-Commerce Insights



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WHAT I DISCOVERED



DATA COLLECTION

Analyzed 100+ printer products from Flipkart, aggregating critical data points on pricing, customer ratings, and brand distribution.



MARKET ANALYSIS

Identified key market trends, brand dominance, and consumer preferences through rigorous competitive analysis.



ACTIONABLE INSIGHTS

Generated data-driven e-commerce insights to help understand value gaps and opportunity areas in the printer segment.

MY TECHNICAL ARSENAL



Python

Primary
Language



BeautifulSoup

HTML
Parsing



Requests

Web
Access



Pandas

Dataframe
s



Regex

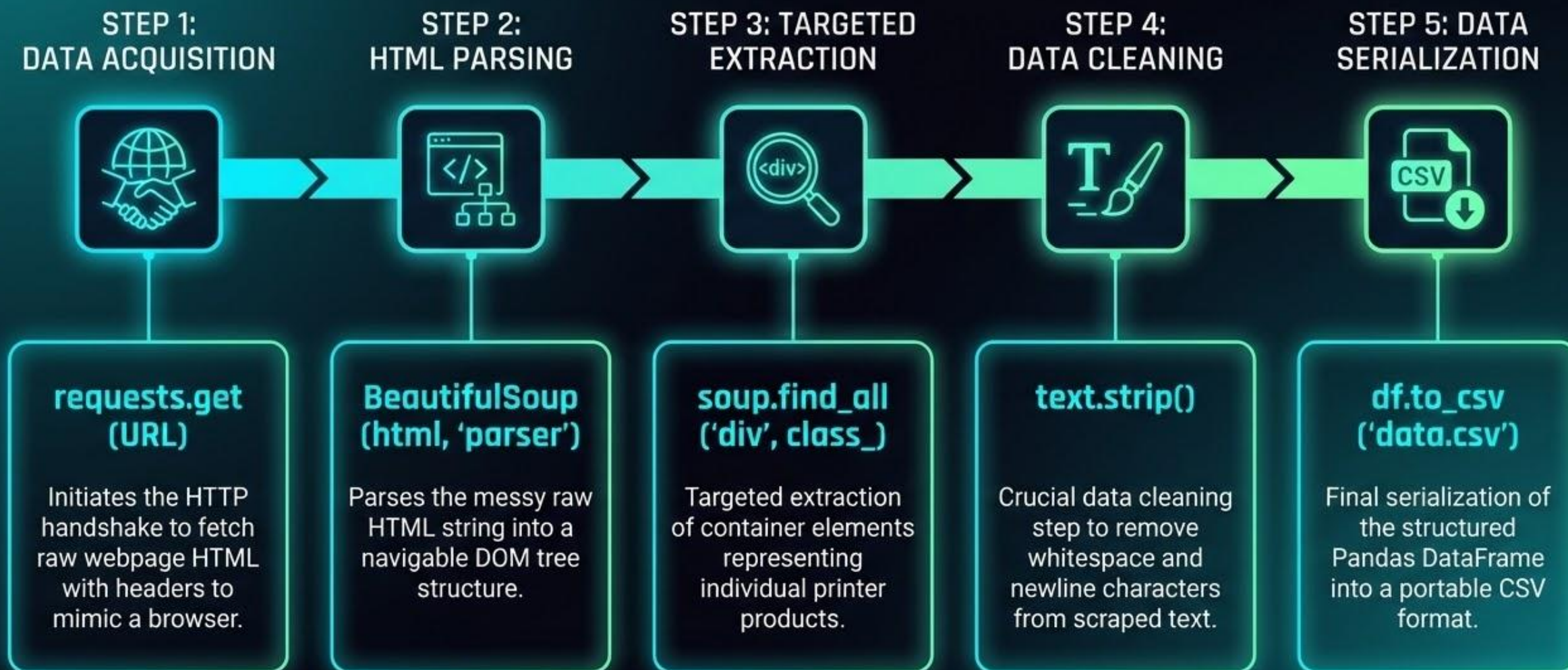
Data
Cleaning

"Leveraging a robust Python ecosystem to transform raw HTML into structured, analytical datasets."

HOW I BUILT IT



PROJECT ROADMAP: CORE FUNCTIONS & WORKFLOW



MARKET DOMINANCE ANALYSIS

BRAND MARKET SHARE



KEY FINDINGS

-  **Oligopoly Detected:** Top 3 brands control 80% of the market listings.
-  **Price Segments:**
Budget: ₹8K-12K | Mid: ₹12K-20K | Pro: ₹20K+
-  **Quality Perception:** Average rating is 4.1/5, indicating high consumer satisfaction across established brands.

MARKET OPPORTUNITIES IDENTIFIED



THE VALUE GAP

There is a scarcity of "high-spec/low-price" products. This creates a clear entry opportunity for disruptor brands to offer premium features at mid-range pricing.



QUALITY PAIN POINTS

Low-rated products consistently suffer from three specific issues: build quality, wireless connectivity failures, and printhead clogging.



BRAND FORTRESS

High trust barriers exist. New entrants must invest heavily in warranty and support to compete with the "Safe Choice" status of HP and Canon.



NON-LINEAR PRICING

Higher specs do not always equal proportionally higher prices. Smart shopping can yield professional features at consumer prices.

| WHAT I LEARNED

01

TECHNICAL MASTERY

Gained deep understanding of DOM structure, HTTP request headers, and handling dynamic web content using Python.

02

DATA INTELLIGENCE

Learned to look beyond raw numbers to find narrative patterns, market gaps, and competitive positioning in datasets.

03

PROBLEM SOLVING

Overcame challenges like irregular HTML structures, missing data fields, and data type conversion errors during parsing.

REAL-WORLD APPLICATIONS

E-Commerce Strategy:

-  Dynamic pricing models based on competitor data.

Retail Analytics:

-  Inventory planning based on popularity and rating trends.

New Brand Entry:

-  Targeting specific feature gaps identified in user reviews.

Academic Growth:

-  A production-ready template for future data science projects.



THANK YOU!

Data-driven insights through intelligent web scraping.

 **Shaik Mohammad Mastan Vali**

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